

HEProAI Activity 3: Machine Learning Cluster Interpretation

Unsupervised Segmentation and Risk Detection Report

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1. Introduction

While Activity 2 utilized rule-based intelligence, Activity 3 employs **Unsupervised Machine Learning (K-Means Clustering)** to identify non-linear patterns within the student dataset. This approach reveals hidden segments, such as students with high academic performance who are simultaneously experiencing extreme burnout.

2. Methodology

The feature vector $X = [APS, WWS, PTMS, CRS]$ was normalized using `StandardScaler` to ensure equal weightage during distance calculation. The K-Means algorithm was applied with $K = 4$, optimizing the Within-Cluster Sum of Squares (WCSS).

3. Cluster Profiling & Analysis

Based on the centroid means identified by the model, the population is segmented as follows:

Cluster	APS	WWS	PTMS	CRS	Persona Name
Cluster 0	83.07	65.33	53.25	42.25	The Career-Undecided Scholar
Cluster 1	73.40	34.44	62.36	66.66	The High-Stress Achiever
Cluster 2	66.16	45.55	46.95	41.66	The Comprehensive Risk Group
Cluster 3	78.42	55.38	82.19	40.00	The Unaligned Power Worker

3.1 Detailed Interpretations

- **Cluster 0 (Purple):** These students are academically elite but lack professional orientation. They represent a "low career clarity" risk despite high grades.
- **Cluster 1 (Blue):** A critical segment identified by ML. They maintain high career readiness but possess the **lowest Wellness Score (34.44)**, indicating severe burnout.
- **Cluster 2 (Green):** This group represents the most significant challenge, scoring lowest in academics and career readiness, requiring 360-degree intervention.
- **Cluster 3 (Yellow):** Characterized by extreme productivity ($PTMS \approx 82$), these students are efficient executors but lack a clear career destination.

4. Comparative Insight: Rule-Based vs. ML

The machine learning model proved more sensitive than the rule-based SRI. For instance, students in Cluster 1 might have been classified as "Green" (Stable) by rules due to high grades, but the ML model correctly flagged them as a distinct "Wellness Risk" group by observing the correlation between their high career focus and declining mental health.